

Volume III. Seventh and Eighth Sections.

§§ 77–78. Modular correspondences and transformation groups.

§ 77. Geometric interpretation of the irrational modular equations as modular correspondences.

We shall make the content of the irrational forms of the modular equations more vivid by a geometric interpretation. First consider the case $n \equiv -1 \pmod{8}$ and put

$$x = f(\omega), \quad y = f_1(\omega), \quad z = f_2(\omega). \quad (1)$$

Then, if x, y, z are regarded as Cartesian coordinates of a point P , a space curve is represented, which we may also express by the two equations

$$x^8 - y^8 - z^8 = 0, \quad xyz = \sqrt{2}, \quad (2)$$

and which we shall call the fundamental curve. This curve is therefore of order 24. It can be mapped onto a plane curve of order 3 if one puts $x^8 = X$, $y^8 = Y$:

$$XY(X - Y) = 16. \quad (3)$$

Putting [§ 75, (1)]

$$\xi = f\left(\frac{c + \delta\omega}{a}\right), \quad \eta = \left(\frac{2}{\delta}\right) f_1\left(\frac{c + \delta\omega}{a}\right), \quad \zeta = \left(\frac{2}{a}\right) f_2\left(\frac{c + \delta\omega}{a}\right), \quad (4)$$

the quantities ξ, η, ζ also satisfy the equations (2), since

$$\left(\frac{2}{a}\right) \left(\frac{2}{\delta}\right) = \left(\frac{2}{n}\right) = \pm 1,$$

and the point Π whose coordinates are ξ, η, ζ therefore also lies on the fundamental curve.

An equation

$$\Phi(x, y, z, \xi, \eta, \zeta) = 0$$

means, when the point P is held fixed, a surface on which Π is to lie, and the intersection of this surface with the fundamental curve gives a certain number of points Π corresponding to the point P . Likewise a certain number of points P correspond to a fixed point Π , and this assignment is called a correspondence on the fundamental curve. If the point P coincides with one of the points Π corresponding to it, we obtain a double point or coincidence of the correspondence.

If equation (5) is not changed when P and Π are interchanged, the correspondence is reciprocal. Such correspondences are given by the equations (8), (9), (10), § 75.

Then by § 75, (3), one has to put

$$\begin{aligned} 2A &= x\xi + (-1)^{(n+1)/8}(y\eta + z\zeta), \\ B &= x\xi y\eta + x\xi z\zeta + (-1)^{(n+1)/8}y\eta z\zeta. \end{aligned} \quad (5)$$

A is of the first degree, and B of the second degree, with respect to the coordinates of each of the two points P and Π .

We shall determine the degree of these correspondences, that is, the number of points Π corresponding to one point P . For this purpose we must multiply the degree of the equation $\Phi = 0$ with respect to ξ, η, ζ by the degree of the fundamental curve, namely by 24, and one obtains:

for $n = 7$,	$m = 24$,
$n = 23$,	$m = 24$,
$n = 31$,	$m = 96$,
$n = 47$,	$m = 48$,
$n = 71$,	$m = 72$,
$n = 15$,	$m = 72$.

In the cases 23, 47, 71, therefore, the degree of the correspondence is equal to the degree of the corresponding modular equation, that is, equal to the number of the transformations in question; in the cases $n = 7, 31, 15$ the degree is three times the degree of the modular equations.

What is the significance of these excess points? They are explained by the fact that in these cases, in which $n \equiv 0, 1 \pmod{3}$, in the function $\Phi_{a,c,\delta}$ [§ 75, (5)], which according to the above equation (5) is equal to a constant, the exponents $h - 2k$ in all terms have the same residue modulo 3, and hence the equation $\Phi = 0$, and likewise the equations (2), remain satisfied when ξ, η, ζ are multiplied by any third root of unity. Thus every three points of the correspondence give the same transformation.

We might also, when x, y, z are determined by (1), have put instead of (4)

$$\xi = f\left(\frac{c + \delta\omega}{a}\right), \quad \eta = \left(\frac{2}{\delta}\right) f_2\left(\frac{c + \delta\omega}{a}\right), \quad \zeta = \left(\frac{2}{a}\right) f_1\left(\frac{c + \delta\omega}{a}\right), \quad (6)$$

and consequently for A, B :

$$\begin{aligned} 2A &= x\xi + (-1)^{(n+1)/8}(y\zeta + z\eta), \\ B &= x\xi y\zeta + x\xi z\eta + (-1)^{(n+1)/8}y\eta z\zeta. \end{aligned} \quad (7)$$

The fundamental curve and the degree of the correspondence would then have remained the same, but the coincidences would have acquired another, and indeed a simpler, meaning. In both cases the coincidences belong to the singular values of the modular functions, in which ω is an imaginary quadratic irrationality, as we shall see more precisely later.

In the case $n \equiv 3 \pmod{8}$ we put

$$\begin{aligned} x &= f^2(\omega), & y &= f_1^2(\omega), & z &= f_2^2(\omega), \\ \xi &= f^2\left(\frac{c + \delta\omega}{a}\right), & \eta &= f_1^2\left(\frac{c + \delta\omega}{a}\right), & \rho &= f_2^2\left(\frac{c + \delta\omega}{a}\right), \end{aligned} \quad (8)$$

and obtain a fundamental curve

$$x^4 - y^4 - z^4 = 0, \quad xyz = 2, \quad (9)$$

which is only of order 12. We then have, further, by § 75,

$$\begin{aligned} 4A &= x\xi - y\eta - z\rho, \\ B &= x\xi y\eta + x\xi z\rho - y\eta z\rho, \end{aligned} \quad (10)$$

and for the degree m of the correspondence one obtains, according to § 75 (13),

$$\begin{array}{l|l} n = 3, & m = 12, \\ n = 11, & m = 12, \\ n = 19, & m = 60. \end{array}$$

Thus again, as in the preceding case, when $n \equiv 0, 1 \pmod{3}$, a number three times too large is obtained.

Finally, if $n \equiv 1 \pmod{4}$, one puts

$$x = f^4(\omega), \quad y = f_1^4(\omega), \quad z = f_2^4(\omega)$$

and obtains a fundamental curve of the sixth degree.

For $n = 5$ the correct number $m = 6$ results.

Eighth Section. The group of the transformation equations and the equation of the fifth degree.

§ 78. The Galois group of the transformation equations for a prime degree.

A more penetrating algebraic study of the transformation equations requires knowledge of their Galois group. Since we have derived the transformation equations from the division equations, whose group is known to us (§ 63), we can likewise form the group of the transformation equations.

The transformation equations arose (§ 65) because the roots of the division equations could be divided into series which were not torn apart by the interchanges of the group of the division equation, but only interchanged among themselves.

To each of these series there corresponds a definite root of a transformation equation, and hence the group of the latter consists of the totality of the interchanges produced by the group of the division equation among the series.

If the transformation degree n is an odd prime number p , this group admits a very elegant representation, suitable for further investigations, and we shall now retain this assumption.

In § 65, (3), the necessary and sufficient condition was already found that two roots of the division equation

$$x_{\mu,\mu'}, \quad x_{\nu,\nu'} \quad (1)$$

belong to the same series R , namely the congruence

$$\mu\nu' - \nu\mu' \equiv 0 \pmod{p}. \quad (2)$$

If now, as is customary in number theory (Gauss, *Disquisitiones arithmeticae*, art. 31), by the symbol

$$\frac{a}{b} \pmod{p}$$

we understand an integer [or residue class $(\text{mod } p)$] which, when multiplied by b , leaves the residue a upon division by p , then, provided that $\mu'\nu'$ are not divisible by p , the congruence (2) defining a series can also be written

$$\frac{\mu}{\mu'} \equiv \frac{\nu}{\nu'} \pmod{p}, \quad (3)$$

and we are therefore naturally led to designate the series R by the value of the ratio

$$\frac{\mu}{\mu'} \equiv z \pmod{p}, \quad (4)$$

which may be congruent to any of the numbers $0, 1, \dots, p-1$, and which is invariant for an entire series. At first the one series remains undesignated in which μ' , and therefore all ν' , are divisible by p ; but this series also fits very well into the general notation if, in case $\mu' \equiv 0 \pmod{p}$, we put

$$\frac{\mu}{\mu'} \equiv \infty \pmod{p}, \quad (5)$$

so that we still obtain a $(p+1)$ st series R_∞ . The totality of the series, whose number is $p+1$, is therefore to be denoted by

$$R_\infty, R_0, R_1, \dots, R_{p-1}. \quad (6)$$

Correspondingly, the associated roots of a transformation equation are to be denoted by

$$v_\infty, v_0, v_1, \dots, v_{p-1}. \quad (7)$$

If, for example, we take the invariant equation (§ 69) as basis, then, according to the determinations of § 68, (10), about the numbers a, c, δ , one has to put

$$v_{\infty} = j(p\omega), \quad v_z = j\left(\frac{z + \omega}{p}\right), \quad (8)$$

and similarly if any other transformation equation is chosen.

With this notation we are able to derive at once the group of the transformation equation from that of the division equation.

We first take as rationality domain the totality of the rational functions of x^2 with rational numerical coefficients.

According to § 63, 3., in this rationality domain the group of the division equation consists of all substitutions by which μ, μ' are transformed into

$$\delta\mu - b\mu', \quad -c\mu + a\mu',$$

where a, b, c, δ are arbitrary integers taken modulo p , whose determinant

$$\Delta = a\delta - bc \quad (9)$$

is not divisible by p , and the number of all these substitutions is

$$p(p-1)(p^2-1) \quad [\S 63, (27)]. \quad (10)$$

From this, however, by the notation (4), (5), the group of the transformation equation is obtained as consisting of all interchanges expressed by the symbol

$$\left(z, \frac{\delta z - b}{-cz + a}\right) \pmod{p}. \quad (11)$$

We denote a substitution (z, z') , if

$$z \equiv \frac{c + \delta z'}{a + bz'} \pmod{p}, \quad (12)$$

similarly as before by

$$\begin{pmatrix} a & b \\ c & \delta \end{pmatrix}, \quad (13)$$

and obtain for the composition of two such substitutions, if

$$z' \equiv \frac{c' + \delta' z''}{a' + b' z''} \pmod{p},$$

the rule

$$\begin{aligned} (z, z')(z', z'') &= \begin{pmatrix} a & b \\ c & \delta \end{pmatrix} \begin{pmatrix} a' & b' \\ c' & \delta' \end{pmatrix} \\ &\equiv \begin{pmatrix} aa' + bc' & ab' + b\delta' \\ ca' + \delta c' & cb' + \delta\delta' \end{pmatrix} \pmod{p}, \end{aligned} \quad (14)$$

in agreement with the rule for the composition of two transformations in § 28. The substitution (11) is therefore to be denoted by

$$\begin{pmatrix} \delta & c \\ b & a \end{pmatrix}.$$

The group formed by all substitutions of this form according to (14) will be denoted by $\mathfrak{L}$ (the group of linear substitutions).

The functions (12) of z' , and hence also the substitutions (13), remain unchanged if the four numbers a, b, c, δ are all multiplied by one and the same factor not divisible by p . Consequently, by (10), the number of these functions, or the degree of the group $\mathfrak{L}$, is

$$p(p^2 - 1), \quad (15)$$

which is also easily found by direct counting.

If in one of the substitutions (13) the four numbers a, b, c, δ are multiplied by a common factor not divisible by p , then the determinant Δ is multiplied by the square of this factor. Hence it is not the determinant Δ which is preserved by this multiplication, but rather its quadratic character, that is, the value of the symbol

$$\left(\frac{\Delta}{p}\right).$$

If two of the substitutions (13) are composed, their determinants multiply, and from this it follows that all substitutions of the group $\mathfrak{L}$ in which Δ is a quadratic residue of p form a group among themselves, which we shall denote by $\mathfrak{L}_0$.

The group $\mathfrak{L}_0$ is a proper divisor of the group $\mathfrak{L}$ of index 2.

If the substitutions of $\mathfrak{L}_0$ are composed with any substitution $(z, \beta z) = \begin{pmatrix} 1 & 0 \\ 0 & \beta^{-1} \end{pmatrix}$, where β and hence also β^{-1} is a quadratic non-residue of p , then the whole group $\mathfrak{L}$ is obtained.

If Δ is a quadratic residue of p , then a common factor to be adjoined to a, b, c, δ can be chosen so that $\Delta = 1 \pmod{p}$, and hence we can represent the group $\mathfrak{L}_0$ also by

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \quad \alpha\delta - \beta\gamma = 1 \pmod{p}. \quad (16)$$

In one of these substitutions the numbers $\alpha, \beta, \gamma, \delta$ are determined modulo p up to the common sign. The degree of the group $\mathfrak{L}_0$ is

$$\frac{1}{2}p(p^2 - 1). \quad (17)$$

But one comes directly to the form (16) if, as group of the division equation, one considers not the group $\mathfrak{U}$, but the group $\mathfrak{B}$ of § 63, that is, if the p th roots of unity are adjoined to the rationality domain, whence the theorem follows:

The group $\mathfrak{L}_0$ is the group of the transformation equation when the p th roots of unity are adjoined to the rationality domain.

For the reduction of the group $\mathfrak{L}$ to the group $\mathfrak{L}_0$, however, the adjunction of a two-valued function is already sufficient; thus the adjunction of the p th roots of unity does too much. The theorems of §§ 63, 65 serve to recognize which irrationality must be adjoined.

In § 63, 3., we saw that the p th root of unity ϱ is rationally representable by the roots of the division equation, and that by a substitution of the group $\mathfrak{U}$ whose determinant is congruent to m , ϱ passes into ϱ^m ; further, in § 65 we proved that by adjoining all roots of a transformation equation the group of the division equation is reduced to the group $\mathfrak{U}_0$, which consists of all substitutions of the form

$$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}, \quad (18)$$

where a is an arbitrary number not divisible by p . The determinants of the substitutions of $\mathfrak{U}_0$ are therefore squares and hence are congruent modulo p to one of the $\frac{1}{2}(p-1)$ quadratic residues of p .

The sum

$$A = \sum_a^a \varrho^a, \quad (19)$$

where a is to run through all quadratic residues of p , therefore remains unchanged under the substitutions of $\mathfrak{U}_0$ and consequently can be expressed rationally by the roots of the transformation

equation. The sum A remains unchanged under the substitutions of the group $\mathfrak{B}$ and hence also under $\mathfrak{L}_0$, while under the substitutions of $\mathfrak{L}$, and hence also of $\mathfrak{U}$, it receives two different values, namely, if b runs through the series of non-residues,

$$A = \sum^a \varrho^a, \quad B = \sum^b \varrho^b.$$

Therefore A is a function belonging to the group $\mathfrak{L}_0$, and by its adjunction $\mathfrak{L}$ is reduced to $\mathfrak{L}_0$.

The values of the sums A, B are known, however (Vol. I, § 179):

$$A = \frac{-1 + \sqrt{(-1)^{(p-1)/2}p}}{2}, \quad B = \frac{-1 - \sqrt{(-1)^{(p-1)/2}p}}{2}.$$

The sign of the root depends on the choice of the root ϱ and can be determined, but this is not relevant here. We therefore have the theorem:

The group of the transformation equation is $\mathfrak{L}_0$ when $\sqrt{(-1)^{(p-1)/2}p}$ is adjoined to the rationality domain.

The group of the invariant equation can also be derived without division of the elliptic functions in the following simple way, though in this way one obtains only the monodromy group, that is, the group in the field of rational functions of $j(\omega)$, without regard to the constants to be adjoined. We restrict ourselves to the case of a prime degree p of the transformation, and put

$$v_c = j\left(\frac{c + \omega}{p}\right), \quad v_\infty = j(p\omega), \quad u = j(\omega), \quad (20)$$

where c is to be taken modulo p .

If a linear substitution

$$S = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

is applied to ω , then $j(\omega)$ remains unchanged and v_c passes into $v_{c'}$. To find c' , one has to seek a second linear substitution

$$S' = \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix}$$

such that

$$\begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix} \begin{pmatrix} p & 0 \\ c' & 1 \end{pmatrix} = \begin{pmatrix} p & 0 \\ c & 1 \end{pmatrix} \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

or

$$\begin{aligned} p\alpha' + c'\beta' &= p\alpha, & \beta' &= p\beta, \\ p\gamma' + c'\delta' &= c\alpha + \gamma, & \delta' &= c\beta + \delta. \end{aligned}$$

From this, by the third equation, $c'\delta' = c\alpha + \gamma$; and with the help of the fourth,

$$c' = \frac{c\alpha + \gamma}{c\beta + \delta} \pmod{p}.$$

If $c\beta + \delta = 0$, then $c' = \infty$; and if $c = \infty$, then $c' = \alpha/\beta$.

In applying this to the determination of c, c' , one may replace $\alpha, \beta, \gamma, \delta$ by congruent numbers a, b, c, d whose determinant $ad - bc$ is a quadratic residue of p , and it can also be shown that in this way every substitution of the group $\mathfrak{L}_0$ can be obtained. Thus every function of v which remains unchanged under the substitutions of the group $\mathfrak{L}_0$ also remains unchanged when a linear substitution is applied to ω , and is consequently a rational function of $j(\omega)$. But as to the nature of the coefficients in this function, this consideration teaches us nothing, and hence $\mathfrak{L}_0$ is recognized only as the monodromy group of the invariant equation.